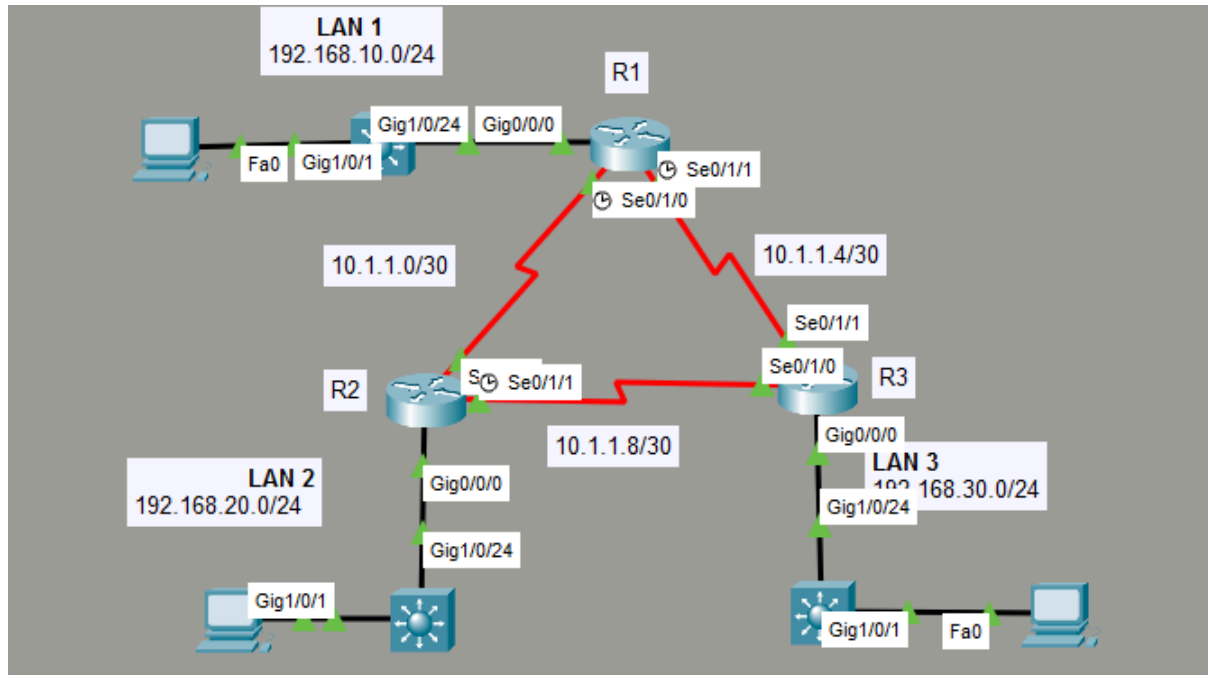


Cel:

Packet Tracer - Point-to-Point Single-Area OSPFv2 Configuration



Addressing Table

Device	Interface	IP Address	Subnet Mask
R1	G0/0/0	192.168.10.1	/24
	S0/1/0	10.1.1.1	/30
	S0/1/1	10.1.1.5	/30
R2	G0/0/0	192.168.20.1	/24
	S0/1/0	10.1.1.2	/30
	S0/1/1	10.1.1.9	/30
R3	G0/0/0	192.168.30.1	/24
	S0/1/0	10.1.1.10	/30
	S0/1/1	10.1.1.6	/30
PC1	NIC	192.168.10.10	/24
PC2	NIC	192.168.20.10	/24
PC3	NIC	192.168.30.10	/24

Objectives

Part 1: Configure Router IDs.

Part 2: Configure Networks for OSPF Routing.

Part 3: Configure Passive Interfaces.

Part 4: Verify OSPF configuration.

Background

In this activity, you will activate OSPF routing using network statements and wildcard masks, configuring OSPF routing on interfaces, and by using network statements quad-zero masks. In addition, you will configure explicit router IDs and passive interfaces.

Instructions

Part 1: Configure router IDs.

- a. Start the OSPF routing process on all three routers. Use process ID **10**.

```
Router(config)# router ospf process-id
```

- b. Use the router-id command to set the OSPF IDs of the three routers as follows

- R1: **1.1.1.1**
- R2: **2.2.2.2**
- R3: **3.3.3.3**

Use the following command:

```
Router(config-router)# router-id rid
```

Part 2: Configure Networks for OSPF Routing

Step 1: Configure networks for OSPF routing using network commands and wildcard masks.

How many statements are required to configure OSPF to route all the networks attached to router R1?

The LAN attached to router R1 has a /24 mask. What is the equivalent of this mask in dotted decimal representation?

Subtract the dotted decimal subnet mask from 255.255.255.255. What is the result?

What is the dotted decimal equivalent of the /30 subnet mask?

Subtract the dotted decimal representation of the /30 mask from 255.255.255.255. What is the result?

- a. Configure the routing process on R1 with the network statements and wildcard masks that are required to activate OSPF routing for all the attached networks. The network statement values should be the network or subnet addresses of the configured networks.

```
Router(config-router)# network network-address wildcard-mask area  
area-id
```

- b. Verify that OSPF has been configured properly by the displaying the running configuration. If you find an error, delete the network statement using the **no** command and reconfigure it.

Step 2: Configure networks for OSPF routing using interface IP addresses and quad-zero masks.

On router R2, configure OSPF using network commands with the IP addresses of the interfaces and quad-zero masks. The syntax of the network command is the same as was used above.

Step 3: Configure OSPF routing on router interfaces

On router R3, configure the required interfaces with OSPF.

Which interfaces on R3 should be configured with OSPF?

Configure each interface using the command syntax shown below:

```
Router(config-if)# ip ospf process-id area area-id
```

Part 3: Configure Passive Interfaces

OSPF will send its protocol traffic out of all interfaces that are participating in the OSPF process. On links that are not configured to other networks, such as LANs, this unnecessary traffic consumes resources. The passive-interface command will prevent the OSPF process from sending unnecessary routing protocol traffic out LAN interfaces.

Which interfaces on R1, R2, and R3 are a LAN interfaces?

Configure the OSPF process on each of the three routers with the **passive-interface** command.

```
Router(config-router)# passive-interface interface
```

Part 4: Verify OSPF Configuration

Use **show** commands to verify the network and passive interface configuration of the OSPF process on each router.

Rozwiązanie:

Part 1:

```
R1>
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 10
R1(config-router)#router-
R1(config-router)#router-id 1.1.1.1
R1(config-router)#|

R2>
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 10
R2(config-router)#router
R2(config-router)#router-id 2.2.2.2
R2(config-router)#|
```

```

R3>en
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 10
R3(config-router)#router-id 3.3.3.3
R3(config-router)#

```

Part 2:

Step 1:

```

R1(config-router)#network 192.168.10.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#network 10.1.1.4 0.0.0.3 area 0
R1(config-router)#do show running-config
R1(config-router)#do show running-config | sect
R1(config-router)#do show running-config | section ospf
router ospf 10
  router-id 1.1.1.1
  log-adjacency-changes
  network 192.168.10.0 0.0.0.255 area 0
  network 10.1.1.0 0.0.0.3 area 0
  network 10.1.1.4 0.0.0.3 area 0
R1(config-router)#

```

Step 2:

```

R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#net
06:07:38: %OSPF-5-ADJCHG: Process 10, Nbr 1.1.1.1 on Serial0/1/0 from
Don
R2(config-router)#network 10.1.1.8 0.0.0.3 area 0
R2(config-router)#network 192.168.20.0 0.0.0.255 area 0
R2(config-router)#do show running-config | include ospf
router ospf 10
R2(config-router)#do show running-config | section ospf
router ospf 10
  router-id 2.2.2.2
  log-adjacency-changes
  network 10.1.1.0 0.0.0.3 area 0
  network 10.1.1.8 0.0.0.3 area 0
  network 192.168.20.0 0.0.0.255 area 0
R2(config-router)#

```

Step:

```

R3(config-router)#int se0
R3(config-router)#int se0/1/1
R3(config-if)#ip ospf 10 area 0
R3(config-if)#
06:10:16: %OSPF-5-ADJCHG: Process 10, Nbr 1.1.1.1 on Serial0/1/1 from LOADING to FULL, Loading
Done

R3(config-if)#int se0/1/0
R3(config-if)#ip ospf 10 area 0
R3(config-if)#
06:10:26: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on Serial0/1/0 from LOADING to FULL, Loading
Done

R3(config-if)#int g0/0/0
R3(config-if)#ip ospf 10 area 0
R3(config-if)#

```

Part 3:

```
R1(config-router)#passi
R1(config-router)#passive-interface gig0/0/0
R1(config-router)#

R2(config-router)#passive-int g0/0/0
R2(config-router)#

R3(config-if)#ip ospf 10 area 0
R3(config-if)#
R3(config-if)#router ospf 10
R3(config-router)#pass
R3(config-router)#passive-interface g0/0/0
R3(config-router)#

R3(config-router)#do show ip ospf
Routing Process "ospf 10" with ID 3.3.3.3
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0

R3(config-router)#do show running
R3(config-router)#do show running-config | section ospf
ip ospf 10 area 0
ip ospf 10 area 0
ip ospf 10 area 0
router ospf 10
router-id 3.3.3.3
log-adjacency-changes
passive-interface GigabitEthernet0/0/0
R3(config-router)#
```